Given that X is thrice as likely to win as Y.

$$A : B \ :: \ 3 : 1. \qquad (1)$$

Y is twice as likely as to win Z

$$B : C :: \ 2 : 1 \qquad (2)$$

From (1) and (2)

$$A : B : C :: \ 6 : 2 : 1$$

$A = 6k, \quad B = 2k, \quad C = k$, where k is proportional constant.

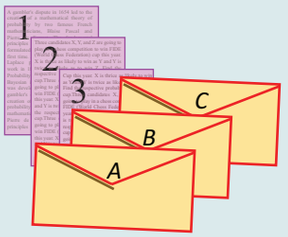
Probability to win the cup by X is $P(A) = \frac{6k}{9k} = \frac{2}{3}$

Probability to win the cup by Y is $P(B) = \frac{2k}{9k} = \frac{2}{9}$ and

Probability to win the cup by Z is $P(C) = \frac{k}{9k} = \frac{1}{9}.$

Example 12.8

Three letters are written to three different persons and addresses on three envelopes are also written. Without looking at the addresses, what is the probability that (i) exactly one letter goes to the right envelopes (ii) none of the letters go into the right envelopes?



Solution

Let A, B, and C denote the envelopes and 1, 2, and 3 denote the corresponding letters.

The different combination of letters put into the envelopes are shown in the table.

Let c_i denote the outcomes of the events.

Let X be the event of putting the letters into the exactly only one right envelopes.

Let Y be the event of putting none of the letters into the right envelope.

Outcomes

Envelope	c_1	c_2	c_3	c_4	c_5	c_6
A	1	1	2	2	3	3
B	2	3	1	3	1	2
C	3	2	3	1	2	1

$S = \{c_1, c_2, c_3, c_4, c_5, c_6\}, \ n(S) = 6$

$X = \{c_2, c_3, c_6\}, \quad n(X) = 3$

$Y = \{c_4, c_5\} \quad n(Y) = 2$

(i) $P(X) = \frac{3}{6} = \frac{1}{2}$ (ii) $P(Y) = \frac{2}{6} = \frac{1}{3}.$

Example 12.9

Let the matrix $M = \begin{bmatrix} x & y \\ z & 1 \end{bmatrix}$. If x, y and z are chosen at random from the set $\{1, 2, 3\}$, and repetition is allowed $(i.e., x = y = z)$, what is the probability that the given matrix M is a singular matrix?

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